

Morris Alper, Ph.D.

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I research creative multimodal machine learning inspired by linguistics—how structures encoded by language provide a tool for interpreting and controlling modalities such as vision and audio, ML as a creativity aid, and innovative methods for low-resource languages.

Education

- Ph.D., Tel Aviv University** 2023 – 2025
Electrical Engineering (Machine Learning)
Advisors: *Dr. Hadar Averbuch-Elor, Dr. Raja Giryes*
Dissertation: *Multimodal Learning for High-Level Semantic Understanding: From Interpretability to Real-World Applications*
- M.Sc. (cum laude), Tel Aviv University** 2019 – 2023
Computer Science
Advisor: *Dr. Hadar Averbuch-Elor*
Thesis: *Learning Human-Human Interactions in Images with Weak Textual Supervision*
GPA: 98 / 100
- B.Sc., Massachusetts Institute of Technology (MIT)** 2012 – 2016
Linguistics and Mathematics (dual major). GPA: 4.9 / 5

Employment History

- Assistant Professor**, University of Miami 2026 – [ongoing]
- Research Analyst (Postdoctoral)**, Carnegie Mellon University 2025 – 2026
Advisor: *Dr. Lori Levin*
Developing machine learning methods for endangered language documentation.
- Research Scientist Intern**, Meta 2024
Developing 3D generative models using weakly-structured image collections.
- Data Science Lead and Lecturer**, Israel Tech Challenge 2019 – 2023
Designed syllabus, wrote, and delivered lectures on data science to industry professionals.
- NLP Data Scientist**, Pipl, Inc. 2019 – 2021
Designed information extraction algorithms for unstructured textual web data.
- NLP Data Scientist**, Reverso-Softissimo 2018 – 2019
Designed NLP algorithms to automatically correct and enrich linguistic data.
- Software Developer**, K Dictionaries 2017 – 2018
Managed API for lexicographic data, and developed algorithms for data enrichment.
- Algorithm Development Intern**, SECOM 2016
Developed algorithms for improved facial recognition.
- Software and Algorithm Developer**, Voiceitt 2015 – 2016
Prototyped speech recognition algorithms and integrated them into the company's API.

Research Publications

* indicates equal contribution.

Peer-Reviewed Conference Papers

1. Yakov Kolani, Maxim Melichov, Cobi Calev, **Morris Alper**. Phonikud: Overcoming Phonetic Under-specification for Hebrew Text-To-Speech. *Interspeech*, 2026.
2. **Morris Alper***, Moran Yanuka*, Raja Giryes, Gašper Beguš. ConlangCrafter: Constructing Languages with a Multi-Hop LLM Pipeline. *ACL*, 2026. (*Oral*)
3. **Morris Alper**, David Novotny, Filippos Kokkinos, Hadar Averbuch-Elor, Tom Monnier. WildCAT3D: Appearance-Aware Multi-View Diffusion in the Wild. *NeurIPS*, 2025.
4. Rachel Mikulinsky*, **Morris Alper***, Shai Gordin, Enrique Jiménez, Yoram Cohen, Hadar Averbuch-Elor. ProtoSnap: Prototype Alignment For Cuneiform Signs. *ICLR*, 2025.
5. Keren Ganon*, **Morris Alper***, Rachel Mikulinsky, Hadar Averbuch-Elor. WAFFLE: Multimodal Floor-plan Understanding in the Wild. *WACV*, 2025. (*Oral*)
6. **Morris Alper** and Hadar Averbuch-Elor. Emergent Visual-Semantic Hierarchies in Image-Text Representations. *ECCV*, 2024. (*Oral*)
7. Assaf Ben-Kish, Moran Yanuka, **Morris Alper**, Raja Giryes, Hadar Averbuch-Elor. Mitigating Open-Vocabulary Caption Hallucinations. *EMNLP*, 2024.
8. Moran Yanuka, **Morris Alper**, Hadar Averbuch-Elor, Raja Giryes. ICC: Quantifying Image Caption Concreteness for Multimodal Dataset Curation. *ACL (Findings)*, 2024.
9. Chen Dudai*, **Morris Alper***, Hana Bezalel, Rana Hanocka, Itai Lang, and Hadar Averbuch-Elor. HaLo-NeRF: Learning Geometry-Guided Semantics for Exploring Unconstrained Photo Collections. *EG*, 2024.
10. **Morris Alper** and Hadar Averbuch-Elor. Kiki or Bouba? Sound Symbolism in Vision-and-Language Models. *NeurIPS*, 2023. (*Spotlight*)
11. **Morris Alper** and Hadar Averbuch-Elor. Learning Human-Human Interactions in Images from Weak Textual Supervision. *ICCV*, 2023.
12. **Morris Alper***, Michael Fiman*, and Hadar Averbuch-Elor. Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding. *CVPR*, 2023.

Workshop and Non-Peer-Reviewed Papers

1. **Morris Alper**, Samopriya Basu, Nathan Bennett, Ryan Kessler, S Verlaïne Ravana, Wendy F. K’ah Skáahluwaa Todd. Towards Low-Resource Text-to-Speech Generation for Indigenous Pacific Northwest Languages: The Case of Haida Xaad Kíl. *ICSNL*, 2025.
2. Shai Gordin, **Morris Alper**, Avital Romach, Luis Saenz Santos, Naama Yochai, Roey Lalazar. CuReD: Deep Learning OCR for Cuneiform Text Editions and Legacy Materials. *ML4AL Workshop*, 2024.
3. **Morris Alper**. Auto-generating Bilingual Dictionaries: Results of the TIAD-2017 Shared Task Baseline Algorithm. *LDK Workshops*, 2017.

Preprints and Under Review

1. Maxim Melichov*, Yakov Kolani*, **Morris Alper**. ReNikud: Audio-Supervised Hebrew Grapheme-to-Phoneme Conversion. 2026.
2. **Morris Alper**, Vasudha Varadarajan, Moran Yanuka, Angelina Wang, Hadar Averbuch-Elor. NAME-SAKES: Probing Identity Memorization in Text-to-Image Models. 2026.
3. **Morris Alper***, Enora Rice*, Bhargav Shaldilya*, Alexis Palmer, Lori Levin. CWoMP: Morpheme Representation Learning for Interlinear Glossing. 2026.
4. Shahaf Pruss*, **Morris Alper***, Hadar Averbuch-Elor. Dynamic Scene Understanding from Vision-Language Representations. 2025.

Teaching Experience

- Computer Vision for Material Culture (Digital Humanities)**, DANES ScHack 2026 Summer School, Turin, Italy July 2026
- Human Language for Artificial Intelligence**, teaching assistant, Carnegie Mellon University Fall 2025
- Digital Signal Processing**, teaching assistant, Tel Aviv University Fall 2023
- Data Science Lead and Lecturer**, Israel Tech Challenge 2019 – 2023
- Designed syllabus spanning fundamentals through advanced topics (deep learning, NLP, computer vision)
 - Lectured on data science and machine learning to diverse industry professionals across multiple cohorts
 - Mentored student capstone projects in technical implementation and communicating results
 - Achieved strong student evaluations (consistently above 6.0 on 7-point scale)

Grants and Funding

Awarded

- Paleography Powered by Machine Learning: Analyzing Cuneiform Documents with Neural Networks** 2023
- LMU-TAU Research Cooperation Program
- Role: Grant co-writer and funded researcher (PIs: Yoram Cohen, Hadar Averbuch-Elor, Enrique Jiménez)

Pending

- Tracing the Signs: Multimodal Foundation Models for Ancient Writing** 2026
- Level I Grant, Humanities and Artificial Intelligence Virtual Institute (HAVI; Schmidt Sciences)
- Role: Co-PI (Lead PI: Ahmad Al-Jallad)

Talks and Presentations

Invited Talks (Selected)

- 2026 **Sound, Script, and Syntax in Multimodal AI**. Technion - Israel Institute of Technology.
- 2026 **Effective Collaboration as Bilingualism: Lessons from AI-Humanities Research**. Archaeology, Ancient History, and AI Workshop, Johns Hopkins University.
- 2026 **ConlangCrafter: Constructing Languages with a Multi-Hop LLM Pipeline**. Workshop on Invented, Constructed, and Emergent Languages in Multi-AI Systems, MIT.
- 2026 **CWoMP: Morpheme Representation Learning for Interlinear Glossing**. NLP Seminar, National Research Council Canada.
- 2026 **Phonikud: Overcoming Phonetic Underspecification for Hebrew Text-To-Speech**. WAVLab Seminar, CMU.
- 2026 **Old Words, New Words: ML for Ancient and Fictional Languages**. Guest Lecture for Linguistics 265: Biological and Artificial Language, UC Berkeley.
- 2025 **Old Words, New Words: ML for Ancient and Fictional Languages**. IBM Israel.
- 2025 **Understanding the Knowledge of Vision-Language Models**. SC Lab Seminar, UC Berkeley.
- 2025 **Emergent Visual-Semantic Hierarchies in Image-Text Representations**. NYC Computer Vision Day 2025.
- 2024 **Understanding the Knowledge of Vision-Language Models**. IMAGINE Lab Seminar, Ecole Nationale des Ponts et Chaussées.
- 2024 **Emergent Visual-Semantic Hierarchies in Image-Text Representations**. Vision & AI Seminar, Weizmann Institute of Science.
- 2024 **Kiki or Bouba? Sound Symbolism in Vision-and-Language Models**. Israel Machine Vision Conference (IMVC).

- 2023 **What is Learned by Vision and Language Models**. School of Electrical and Computer Engineering, Ben Gurion University.
- 2023 **What is Learned by Vision and Language Models**. Shevah Mofet High School Linguistics Club.
- 2023 **Is BERT Blind? Exploring the Effect of Vision-and-Language Pretraining on Visual Language Understanding**. Vision Seminar, Hebrew University of Jerusalem.
- 2023 **“Deplomantics”: Deep learning for Automatic Analysis of Cuneiform Texts**. First Meeting of the Digital Ancient Near Eastern Studies Network (DANES).
- 2022 **“Deplomantics”: Deep learning for Automatic Analysis of Cuneiform Texts**. LMU-TAU Workshop for Early-Career Researchers.
- 2017 **Auto generating bilingual dictionaries**. eLex 2017 conference, Leiden, the Netherlands.

Industry Workshops (delivered through Israel Tech Challenge)

- 2021–2022 **NLP Workshop Series**. AT&T Israel.
- 2020 **Cutting-Edge NLP With Transformers and BERT**. PayPal Israel.

Press Coverage

- 2026 ConlangCrafter featured in Science, IEEE Spectrum, University of Miami News, and Sun Sentinel
- 2025 ProtoSnap featured in Cornell News
- 2024 Kiki-Bouba sound symbolism work featured in Haaretz (Hebrew)
- 2024 Radio interview for Kan Tarbut program on VLM sound symbolism research (46:30-60:00, Hebrew language) [audio]

Recorded Presentations

- 2024 IMVC 2024 presentation on “Kiki or Bouba?” NeurIPS 2023 paper [video]
- 2023 “Deplomantics” presentation at DANES [video]
- 2017 “Auto generating bilingual dictionaries” at eLex [video]

Miscellaneous Experience

Awards and Achievements

- 2011 **International Linguistics Olympiad**, **first place internationally** (both individual and team rounds), best solution (problem 2), Zhurinsky Memorial Prize.
- 2009 **International Linguistics Olympiad**, honorable mention (individual round), first place in team round.
- 2008 **International Linguistics Olympiad**, tied for 6th place individual round (silver medal).

Peer Review Service

- 2026 ACL ARR (Jan, Mar, May), CoLM, Humanities and Social Sciences Communications [ongoing]
- 2025 SIGGRAPH, NeurIPS (*Top Reviewer*), ICLR, ACL ARR
- 2024 COLM, NeurIPS, ACML, ICLR

Languages

English (native), **Hebrew** (professional), **French** (upper intermediate)